

The Rise of Generative AI Increases Job Insecurity and Behavioral Adaptation Traps: Structural Evidence from Heterogeneous Volatility and SVAR Models

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Abstract. The rapid commercialization and market-wide salience of generative artificial intelligence (GenAI) have introduced profound structural demands across global economic systems, yet macro-organizational psychology research has focused heavily on micro-level workplace settings, leaving the dynamic, high-frequency transmission boundaries of technological anxiety unexamined. Grounded in the job demands-resources (JD-R) model and Hofstede’s cultural dimensions theory, this study deployed a multi-model time-series framework, incorporating Structural Vector Autoregressions (SVARs), split-sample Autoregressive Distributed Lag (ARDL) Bounds testing, and out-of-sample machine learning validation, to capture the systemic impacts of exogenous technology shocks on workforce job insecurity, professional burnout, and voluntary turnover intentions (Job Quitting) within two contrasting cultural architectures: the United States and Vietnam. Formal Chow structural stability tests validated a major post-November 2022 regime shift globally following the public launch of ChatGPT. In the individualistic United States system, generative shocks operated as unmitigated, unidirectional job demands ($\Delta\text{GenAI Shock} \rightarrow \Delta\text{Job Insecurity}$, $p = 0.004$), triggering an immediate, intense resource depletion process that accounted for approximately 38.77% of the synchronized variance across emotional exhaustion and market attrition. Conversely, the collectivist Vietnamese architecture exposed a powerful cultural cushioning mechanism, where strong group networks and institutional protection expectations acted as an active social buffer, downscaling long-run shock elasticities ($\beta_{\text{post}} = +0.473$, $p < 0.001$) and shifting behavioral trajectories toward agentic, defensive information-seeking loops. Out-of-sample Random Forest simulations validated the structural robustness and directional predictive power of these vectors. These findings expand the JD-R framework to macro-market frequencies, highlighting cultural architecture as a critical institutional resource that dictates occupational public health in the automation era.

Keywords: Generative artificial intelligence, job insecurity, burnout, job demands-resources model, cultural buffer, high-frequency macro-data.

1 Introduction

The widespread public commercialization of generative artificial intelligence (GenAI) following the launch of large language models in late 2022 represents one of the most disruptive macroeconomic and evolutionary turning points in the modern history of work [1, 2]. Unlike historical waves of automation that primarily neutralized routine manual and blue-collar labor [3], generative technologies have directly penetrated cognitive, professional, and creative occupational spaces, domains long considered fundamentally insulated by human intellectual property [4]. While economists have extensively modeled the long-term structural adjustments of this shift through the lenses of labor replacement, wage polarization, and productivity optimization [1], a critical, high-frequency human cost remains heavily under-theorized: the psychological, immediate, and systemic erosion of employee welfare across global labor markets.

Historically, organizational psychology has addressed technological change through static, micro-level frameworks, examining how individual workers process automation threats over localized corporate restructurings [5]. However, because GenAI operates as a diffuse macro-phenomenon—transcending specific firm boundaries and maintaining a constant presence in public media—it introduces an ubiquitous, systemic cognitive demand that requires next-generation dynamic analysis [6]. Grounded in the foundational concepts of the job demands-resources (JD-R) framework [7, 8], a major technological leap of this scale acts as an intense, unmitigated structural job demand. When the market-wide salience of automation achieves permanent prominence, it forces immediate cognitive adaptation and presents an ongoing threat to professional identity. According to the JD-R health impairment process, prolonged exposure to such demands without sufficient supporting mechanisms rapidly exhausts human energy reserves [8, 9], leading from chronic subjective job insecurity directly into professional burnout. When cognitive and emotional resources are completely depleted, employees invoke behavioral withdrawal strategies as a final preservation mechanism, triggering structural attrition across the aggregate economic system [10].

Crucially, the speed, intensity, and ultimate behavioral manifestation of this resource depletion pathway do not occur within an institutional vacuum; rather, they are fundamentally regulated by national cultural architectures [11]. By synthesizing the dynamic properties of the JD-R model with Hofstede’s cultural dimensions, this study investigates how underlying cultural configurations alter the transmission vectors of technological panic. We contrast the United States, a hyper-individualistic society characterized by decentralized social security nets, highly fluid employment configurations, and an institutionalized “at-will” labor paradigm [11], against Vietnam, a deeply collectivist, emerging East Asian market sustained by strong group mutual obligations, dense informal social cushioning networks, and structural corporate expectations of employment stability [12]. Within individualistic contexts, macro-technical stress is processed through an isolated lens of personal exposure, creating a volatile, frictionless transmission path where technological innovations rapidly trigger workplace panic. In contrast, within collectivist economies, group resilience networks, collaborative so-

cial support structures, and deep-seated institutional loyalties are hypothesized to act as a powerful psychological and structural “buffer”, intercepting, neutralizing and slowing down the systemic transmission of technological strain [11].

To capture these continuous, non-linear, and structural feedback dynamics over time, traditional cross-sectional surveys or long-lagged annual panels are analytically insufficient. Such micro-data approaches systematically fail to capture the high-frequency conversational momentum, viral anxiety spikes, and out-of-sample predictive robustness that define modern technological adoption waves. To bridge this methodological gap, this study constructs a unique, high-frequency macroeconomic data infrastructure spanning a multi-year timeframe across both national contexts. Leveraging digitized search indices and granular behavioral sentiment vectors, we monitor the weekly interactions between generative automation signals, localized job instability discussions, professional burnout indicators, and voluntary exit intentions (Job Quitting).

Methodologically, our empirical design executes a rigorous multi-stage structural timeline. First, we implement Structural Vector Autoregressions (SVARs) to map the short-run innovations and identify unit-free information variance splits through data-driven Cholesky ordering frameworks. Second, we utilize split-sample Autoregressive Distributed Lag (ARDL) Bounds testing coupled with formal Chow structural break diagnostics to formally evaluate parameter stability across the historical boundary marking the public release of ChatGPT. Third, we protect these inferences against linear misspecification and statistical overfitting by validating our forecasting trajectories through non-linear machine learning out-of-sample models (Random Forest).

By executing this empirical pipeline, this research offers substantial theoretical and practical contributions. Mechanically, it elevates the micro-organizational principles of the JD-R model to dynamic macro-market frequencies, demonstrating that high-frequency technological volatility can systematically drain public health assets. Contextually, it advances cross-cultural psychology by validating national culture not as a passive background variable, but as an active, structural moderator that dictates the transmission speeds of structural macroeconomic shock vectors. Practically, our findings provide a real-time predictive dashboard for corporate leaders, occupational public health officials, and national policy architects seeking to balance the financial acceleration of artificial intelligence integration with the sustainable preservation of global human cognitive capital.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical integration of the JD-R framework, cultural dimensions, and technological transition economics, establishing our core analytical hypotheses. Section 3 details the data collection pipelines, high-frequency index construction metrics, and the multi-model econometric estimation design. Section 4 provides the full empirical results, structural break diagnostics, and predictive machine learning benchmarking. Finally, Section 5 presents our summary conclusions, deep practical policy implications, and future research coordinates.

2 Theoretical Background and Hypothesis Development

The macroeconomic transmission of disruptive technological shocks into aggregate labor market frictions has historically been examined through a structural, production-side displacement lens. However, cognitive automation induced by the diffusion of GenAI demands a multi-theoretical framework that harmonizes macro-level econometrics with organizational behavior and stress theories. This study structuralizes these dynamics by integrating transactional stress mechanisms, resource conservation processes, behavioral entrapment, and cross-national cultural architectures to map how aggregate technological disruptions propagate into psychological labor anomalies over time.

2.1 Theoretical Background

The Transactional Stress Model and Dynamic Appraisal Processes

At a fundamental level, job insecurity reflects a subjectively appraised, future-oriented threat concerning potential involuntary job loss [13]. While traditional occupational health literature routinely positions job insecurity as a static predictor and psychological or physical health as an outcome [13], this historical focus represents an artificial truncation of psychological events based on where in the flow of macro-level cycles one chooses to begin and end the analysis [14]. The transactional stress model deeply criticizes this compartmentalization, conceptualizing stress-related processes as part of a dynamic, reciprocal system over time [14].

Under this meta-theoretical perspective, individuals' reactions to an initial triggering condition do not merely dissipate; instead, they can cause or worsen that condition, creating cascading, self-reinforcing threats to functional health and systemic well-being in the long term [15]. Although depictions of transactional stress place strains as outcomes for illustrative convenience, prolonged exposure to perceived vulnerability acts as a major cause of subsequent threat appraisal. Poor or declining subjective safety triggers a heightened and sustained psychophysiological response with profound repercussions for collective workforce stability [16, 17]. Conversely, maintaining paid work and sustaining economic viability are predicated on the continuous preservation of functional capacities over time [18]. The transactional stress model thus provides a rigorous foundation for conceptualizing these dynamic relations as reciprocal feedback loops within an evolving macroeconomic environment.

Conservation of Resources and Job Demands-Resources Theories

To articulate the specific rationale of why and how technology-induced strains transform into persistent behavioral withdrawal, we supplement the transactional framework with the Conservation of Resources (COR) Theory [10] and the JD-R Model [8]. COR theory assumes that individuals are inherently motivated to accumulate, protect, and foster personal, social, and energetic resources. Psychological stress occurs when these core resources are threatened with loss, or when

individuals invest substantial resources without achieving anticipated complementary returns [10].

When integrated with the JD-R model, environmental stressors are bifurcated into Job Demands (physical, social, or organizational aspects that require sustained cognitive and emotional effort) and Job Resources [8]. A severe, structural escalation in job demands, unmatched by a corresponding expansion in coping resources, activates a destructive health impairment process [8]. Under this dual lens, negative environmental shocks or highly volatile occupational conditions threaten individuals' perceived work ability over time [19]. To counter this resource depletion, individuals expend finite cognitive energy to execute protective actions. However, if the external demand remains aggressively unyielding, this continuous expenditure breeds profound psychological exhaustion and systemic burnout, ultimately forcing agents to engage in defensive, permanent behavioral withdrawal to prevent total resource bankruptcy.

Behavioral Trap Theory While resource depletion theories explain the internal erosion of functional capacity, *Behavioral Trap Theory* explains the structural locking mechanisms that govern collective coping failure [20, 21]. A behavioral trap emerges when an agent is induced to enter a specific course of action by immediate, short-term positive reinforcement, but as the trajectory progresses, it shifts into severe, long-term disadvantageous consequences from which exit is highly costly or prohibited [21]. In organizational ecology, when individuals face an overwhelming threat to their competitive advantage, their immediate secondary appraisal prompts them to initiate proactive adaptation or structural job crafting to integrate the disruption into their routines. However, if the velocity of the environmental stressor systematically outpaces human adaptability, this proactive response backfires. The immediate reinforcements of marginal adaptation blind the system to the compounding long-term resource deficits, entrapping the collective workforce in a vicious cycle where intensified effort accelerates systemic depletion.

Hofstede's Cultural Dimensions Theory Macroeconomic disruptions do not propagate through an institutional or cultural vacuum; rather, threat-to-threat relations change in intensity across national boundaries based on prevailing socio-cultural architectures. To structuralize how cross-country environments moderate the velocity and severity of behavioral traps, we leverage *Hofstede's Cultural Dimensions Theory* [11]. We focus specifically on the configuration of the *Individualism-Collectivism* and *Uncertainty Avoidance* axes. Individualistic cultures emphasize independent self-construals and personal career accountability, meaning that the onus of structural adaptation falls entirely on the individual. Collectivist cultures, conversely, are anchored in interdependent networks where organizations, relational systems, and institutional frameworks serve as collective safety valves and protective buffers [11]. Furthermore, societies with low uncertainty avoidance possess a flexible tolerance for ambiguity and situa-

tional precarity, altering how raw information shocks are absorbed and subjective evaluations of environmental threat are updated over time.

2.2 Hypothesis Development

Generative AI Shocks and Aggregate Job Insecurity Applying the primary appraisal phase of the transactional stress model [14], when a macro-level technological shock enters the labor market, its transmission is contingent upon how its core features are collectively evaluated. Traditional physical automation primarily targeted routine, manual task profiles, leaving high-level cognitive domains insulated [6]. Conversely, GenAI displays advanced capabilities in analytical reasoning, programming, and abstract synthesis, directly encroaching upon non-routine, white-collar knowledge domains. Consequently, the aggregate white-collar workforce appraises the diffusion of GenAI not as a benign tool, but as an existential threat to their core competitive advantage.

This pervasive anxiety manifests as a structural shift in aggregate job insecurity. Because individuals generally view macrolevel technological displacement as highly uncontrollable [22], the persistent perception of threat triggers a heightened and sustained psychophysiological response across the workforce [16]. Rather than acting as a flat, linear stressor, intense and volatile macro-level technological exposures induce significant conditional variance clustering in collective sentiment. The uncertainty surrounding algorithmic capability alters subjective evaluations, shifting the aggregate labor market from states of stability to intense psychological volatility. Accordingly, we hypothesize:

Hypothesis 1 (H1): *Macro-level Generative AI shocks induce significant short-to-medium-run structural shifts in aggregate job insecurity, characterized by high psychological volatility clustering.*

Proactive Adaptation Backfires and the Psychological Rumination Lag Once a primary threat is established, the combination of transactional stress mechanisms [14] and the JD-R health impairment pathway [8] predicts that macro-level AI shocks drive cumulative workforce burnout and exit behaviors. Following secondary appraisal, workers attempt to cope with technology-induced job insecurity by engaging in proactive adaptation, altering task boundaries to integrate the “AI-Coworker” into daily workflows. However, as recent empirical literature demonstrates, when proactive adaptation backfires, it yields unintended consequences of job crafting to integrate AI, leading to severe resource depletion [23]. Under *Behavioral Trap Theory* [21], however, when the diffusion velocity of technology systematically outpaces human cognitive and neurological plasticity, this proactive coping mechanism backfires.

Knowledge workers expend finite emotional and cognitive resources to maintain complementarity with accelerating algorithmic capabilities [6]. This continuous expenditure without structural replenishment triggers severe, systemic resource depletion [10], transforming proactive upskilling into a structural engine for aggregate emotional exhaustion and collective burnout. Crucially, this

threat-to-threat transmission is governed by a strict temporal constraint. Structural shocks in AI volatility do not instantaneously materialize into psychological breakdown or immediate market exit. Instead, the system operates subject to a non-instantaneous *psychological rumination lag*. Workers experience an extended window of cognitive processing, trial-and-error adaptation, and cumulative resource erosion before the system reaches an inflection point toward burnout or triggers the ultimate defensive withdrawal mechanism: job quitting and permanent labor market exit. Thus, we predict:

Hypothesis 2 (H2): *Generative AI shocks structurally drive aggregate workforce burnout and job quitting behaviors over time, subject to a non-instantaneous psychological rumination lag.*

The Moderating Role of Cultural and Institutional Asymmetry A full specification of a dynamic macroeconomic model requires theorizing whether these threat-to-threat relations exhibit non-linear changes in intensity over time. Recent behavioral studies highlight that the psychological impact of AI-induced job insecurity is rarely static, often following a non-linear trajectory from initial despair to eventual hope concerning innovative behavior and adaptation [24]. We argue that this non-linear dynamic is heavily moderated by cross-national cultural architectures following major technological turning points. Following *Regime Shift Theory* and *Cultural Dimensions Theory* [11], the severity and permanence of the behavioral adaptation trap are heavily moderated by cross-national cultural architectures.

In a highly individualistic labor market with low institutional safety nets, such as the United States, the responsibility for technological adaptation is placed entirely on the individual. Driven by individualistic preservation, collective stress experiences high psychological persistence; the independent drive for job crafting backfires intensely, lock-stepping the advanced workforce into a chronic, permanent-memory behavioral trap where technological exposure continuously intensifies workforce burnout.

Conversely, emerging markets like Vietnam operate under deeply embedded collectivist structures and a high, flexible tolerance for situational ambiguity [11]. In this macroeconomic regime, relational networks and collective institutional safety valves shield individual workers from unmitigated technological stress. Rather than forcing immediate, isolated pressure onto individual careers, collectivist organizational dynamics absorb the friction through shared adjustments and gradual assimilation. Consequently, while the initial public turning point of technology may provoke widespread acute anxiety due to media-driven salience, the collectivist buffer effectively dampens long-term cognitive strain. Once the disruption normalizes, the aggregate behavioral friction undergoes a rapid, transient stabilization, preventing chronic adaptation traps. Accordingly, we hypothesize:

Hypothesis 3 (H3): *The macroeconomic transmission mechanism of GenAI shocks is structurally altered following major public technological turning points, manifesting as a highly persistent behavioral adaptation trap in individualistic*

advanced markets (US) and a dampened, non-persistent adjustment in collectivist emerging markets (Vietnam).

3 Method

To evaluate the dynamic macroeconomic transmission of Generative AI shocks into collective labor market anxieties and subsequent withdrawal behaviors, we construct a high-frequency empirical modeling framework. Rather than tracking isolated static samples, our approach reconstructs continuous market-level behavioral regimes by integrating localized digital trace indices with structural macroeconomic series.

3.1 Data Sources and Spatial-Temporal Mapping

The empirical environment is grounded in continuous weekly observations spanning from January 2022 through March 2026, capturing the complete structural transition from the pre-ChatGPT public baseline to the highly commercialized landscape of large language models (LLMs). The framework runs parallel systems for two structurally and culturally distinct economies: the United States (individualistic landscape) and Vietnam (collectivist landscape).

Macroeconomic Control Indicators To isolate technological psychological strains from overarching economic or business cycles, we capture three country-specific monthly macroeconomic series, mapped to a weekly frequency using linear interpolation to match the behavioral timeline. These indicators reflect the baseline structural constraints of each market:

- **Index of Industrial Production (IIP)** (IIP_{sa}): Seasonally adjusted to track real-sector operational capacity and manufacturing contractions.
- **Inflation** ($INFL_{sa}$): Derived from the seasonally adjusted Consumer Price Index (CPI) to capture purchasing power erosion.
- **Bilateral Exchange Rate (EXR)**: Fixed against the United States Dollar (USD) for the Vietnamese system and against a basket of major currencies for the US system to control for external macro-shocks.

Digital Trace Extraction and Keyword Integration To operationalize macro-level psychological strains and behavioral withdrawal, we utilize the Search Volume Index (SVI). It is crucial to delineate the construct validity of this approach. While traditional organizational psychology relies on micro-level psychometric surveys to diagnose individual clinical states, SVI captures a fundamentally different construct: *collective attention* and macro-behavioral momentum. Following Heiberger (2015), aggregate search volumes serve as robust, non-reactive indicators of shifts in collective attention. Recent systematic reviews further confirm that SVI acts as a powerful proxy for large-scale crowd

behavior, demonstrating strong positive correlations with market volatility and significantly improving forecasting models over traditional metrics [25]. Furthermore, capturing this data at a high frequency provides precise insights into large-scale collective decision-making that low-frequency annual surveys systematically miss [25]. Therefore, in this study, SVI is not deployed as a diagnostic tool for individual clinical burnout, but rather as an aggregate barometer for the market-wide salience of technological anxiety and occupational exit intentions.

Aggregate psychological and behavioral expressions are captured via high-frequency internet search footprints retrieved from the Google Trends API. This data records the weekly SVI for localized query networks, serving as a non-reactive behavioral barometer. For each target construct, we utilize a primary seed query to isolate the top co-occurring semantic extensions within each geographic region, purging irrelevant contextual anomalies.

Let $SVI_{c,k}(t)$ denote the raw SVI for construct c in country k at week t . To extract the net, macro-insulated psychological signal ($\hat{\epsilon}_{c,k,t}$), we pass the raw log-transformed indices through an ordinary least squares (OLS) baseline filtering matrix:

$$\ln(SVI_{c,k}(t) + 1) = \alpha_0 + \gamma_1 IIP_{sa,t} + \gamma_2 INFL_{sa,t} + \gamma_3 EXR_t + \epsilon_{c,k,t}, \quad (1)$$

where the estimated surplus $\hat{\epsilon}_{c,k,t}$ isolates pure technology-induced shifts. The resulting macro-insulated variables are operationalized as follows:

1. **GenAI Shock** ($GenAI_{Shock}$): The aggregated macro-insulated index tracking core search queries for LLMs and commercialized generative platforms.
2. **Job Insecurity** ($Job_{Insecurity}$): The filtered search cluster isolating mass layoffs, collective downsizing, and corporate employment precarity inquiries.
3. **Workforce Burnout** ($Burnout$): The insulated index (l_MDI) tracking behavioral signatures of chronic emotional exhaustion and severe workplace cognitive overload.
4. **Job Quitting Intentions** ($Job_{Quitting}$): The structural index (l_ESCAPE) mapping digital traces of active career evasion, resignation protocols, and acute job quitting inquiries.

3.2 Preliminary Analyses and Data Diagnostics (EDA)

Prior to specifying the multivariate structural equations, we execute continuous univariate and bivariate diagnostics to expose the fundamental mathematical, distribution, and temporal properties of the macro-insulated residuals. Following the methodological standards of the *Journal of Applied Psychology* for longitudinal and high-frequency designs, checking these empirical properties is critical to ensure proper model identification, evaluate distributional non-normality, and interpret baseline linear connections before introducing structural lag restrictions.

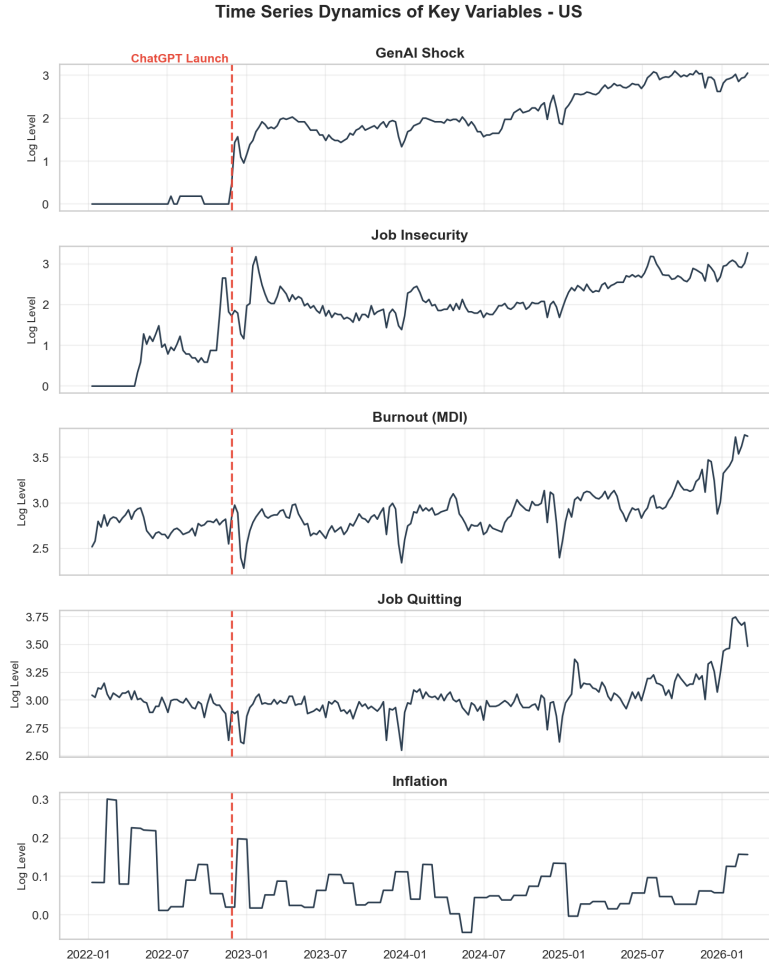


Fig. 1: High-Frequency Weekly Behavioral Trajectories for the United States Regime (Jan 2022 – Mar 2026). The vertical dashed line highlights the exogenous launch of ChatGPT in November 2022.

The macro-insulated behavioral trajectories for the United States system, presented in Figure 1, exhibit severe, permanent non-linear level shifts immediately following the exogenous emergence of commercialized LLMs. Prior to November 2022, l_SHOCK_Index operates on a flat, homoskedastic baseline near zero. The commercialization milestone triggers an immediate, vertical cliff-like escalation in technological exposure, which is instantly shadowed by a highly synchronized upward trend in aggregate $l_VOL_Base_Index$.

Crucially, in structural validation of the resource depletion timeline derived from the JD-R framework, the escalation in workforce l_MDI_Index (Burnout)

and l_ESCAPE_Index (Job Quitting) does not occur simultaneously with the initial technology shock wave. Instead, the US graph reveals a distinct, visible tracking lag of approximately 4 to 8 weeks before l_MDI_Index and l_ESCAPE_Index undergo a permanent outward shift. This structural latency illustrates that the behavioral transmission mechanism operates through historical cumulative strain rather than instantaneous physical reactions, justifying the contemporary zero-restrictions embedded within our subsequent SVAR matrix.

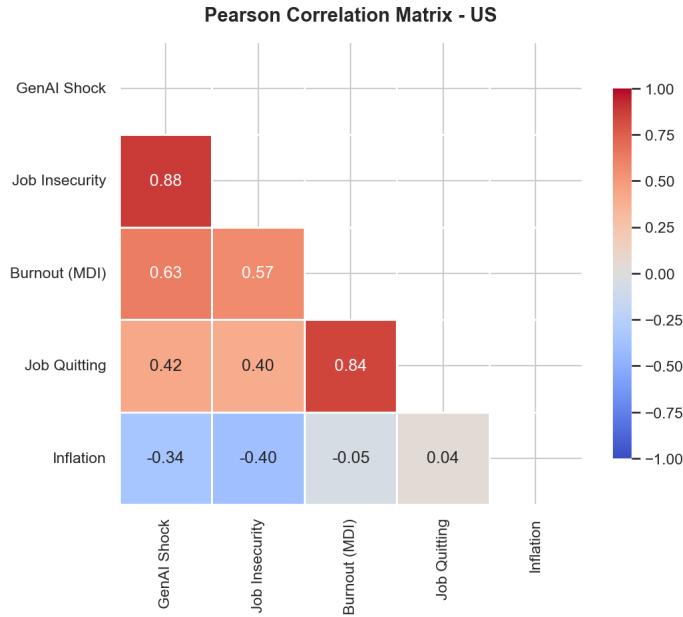


Fig.2: Pearson Correlation Matrix of Macro-Insulated Variables within the United States Labor Architecture.

Figure 2 formalizes the baseline linear landscape for the United States, exposing a highly integrated and dense psychological distress network. The linear connectivity between generative technology exposure and workforce anxiety is exceptionally severe, with l_SHOCK_Index exhibiting an overwhelming positive correlation with $l_VOL_Base_Index$ ($r = 0.88$). This proves that market-level awareness of generative automation is almost perfectly synchronized with collective employment precarity. Furthermore, the US system reveals that l_MDI_Index shares a massive linear overlap with both $l_VOL_Base_Index$ ($r = 0.57$) and l_ESCAPE_Index ($r = 0.84$). This critical baseline confirms that inside fluid, individualistic labor architectures, technological shocks operate as highly systemic threats, where employment fear seamlessly bleeds into emotional exhaustion and immediate market withdrawal intentions.

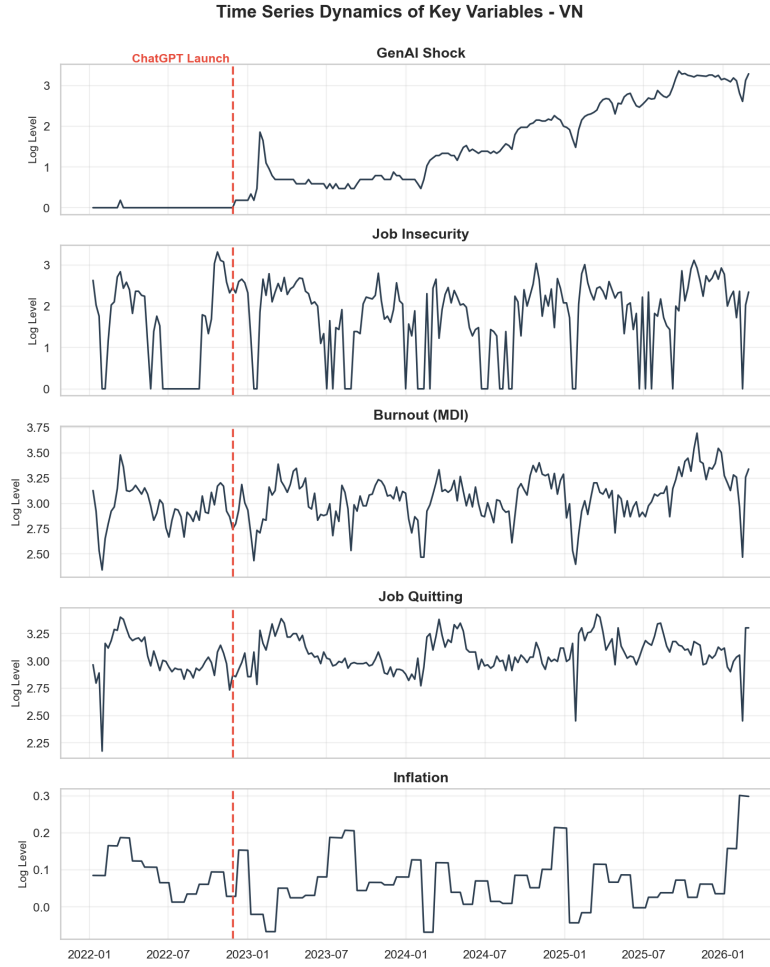


Fig. 3: High-Frequency Weekly Behavioral Trajectories for the Vietnam Regime (Jan 2022 – Mar 2026). The vertical dashed line highlights the exogenous launch of ChatGPT in November 2022.

In stark contrast, the Vietnamese behavioral trajectories illustrated in Figure 3 display a highly distinct, mean-reverting morphology. While the l_SHOCK_Index trajectory in Vietnam experiences an analogous outward jump post-November 2022, reflecting the borderless salience of global technological developments, the subsequent pass-through to labor anxieties behaves in a fundamentally stationary manner. Following the initial technological shockwave, collective $l_VOL_Base_Index$ and l_MDI_Index in Vietnam experience transient, short-lived spikes that rapidly decay back toward the historical equilibrium. This visual patterns suggest a

cushioned, temporary disruption rather than the permanent behavioral traps observed in individualistic markets.

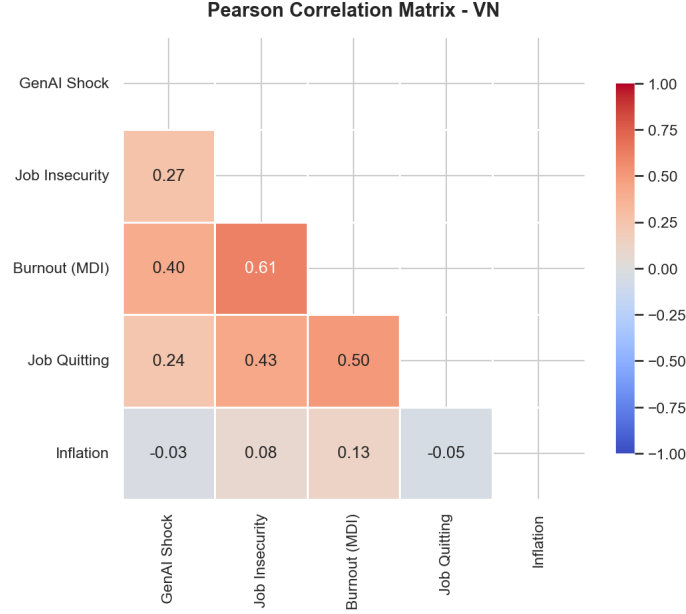


Fig. 4: Pearson Correlation Matrix of Macro-Insulated Variables within the Vietnamese Collectivist Labor Architecture.

The decoupled nature of this emerging market system is mathematically confirmed by the Vietnamese correlation matrix presented in Figure 4. The direct linear pass-through from l_SHOCK_Index to $l_VOL_Base_Index$ is remarkably weak ($r = 0.27$), suggesting that the initial technological shock waves fail to translate directly into pervasive employment panic within a collectivist labor landscape. While the localized correlation between $l_VOL_Base_Index$ and l_MDI_Index remains structurally active ($r = 0.61$), the critical structural link driving ultimate workforce flight—the association between l_ESCAPE_Index and the primary technology shock—is fundamentally diluted ($r = 0.24$). This baseline insulation suggests that institutional and cultural buffering mechanisms in Vietnam successfully break down the direct linear transmission from technological disruption to permanent labor market exit.

To formally secure these visual observations through rigorous mathematical diagnostics, Table 1 reports the finalized empirical time-series parameters.

Table 1: Descriptive Statistics and Joint Integration Diagnostics

	Variable	Mean	S.D.	ADF Level	KPSS Level	ADF Δ^1	Order
US	<i>l_SHOCK_Index</i>	1.715	1.001	0.6678	0.0100	0.0000***	<i>I</i> (1)
	<i>l_VOL_Base_Index</i>	1.896	0.793	0.1674	0.0100	0.0000***	<i>I</i> (1)
	<i>l_MDI_Index</i>	2.890	0.229	0.6925	0.0100	0.0000***	<i>I</i> (1)
	<i>l_LEARN_Index</i>	3.022	0.171	0.9322	0.0100	0.0000***	<i>I</i> (1)
	<i>l_ESCAPE_Index</i>	3.022	0.171	0.7420	0.0100	0.0000***	<i>I</i> (1)
	<i>l_INFL_sa</i>	0.071	0.063	0.0001***	0.0377**	—	<i>I</i> (0)
	<i>l_IIP_sa</i>	0.3988	0.0449	0.3988	0.0449	0.0000***	<i>I</i> (1)
	<i>l_EXR</i>	0.0691	0.0632	0.0691	0.1000	0.0000***	<i>I</i> (1)
VN	<i>l_SHOCK_Index</i>	1.307	1.107	0.9092	0.0100	0.0000***	<i>I</i> (1)
	<i>l_VOL_Base_Index</i>	1.734	0.957	0.0000***	0.1000	—	<i>I</i> (0)
	<i>l_MDI_Index</i>	3.039	0.225	0.0000***	0.0321**	—	<i>I</i> (0)
	<i>l_LEARN_Index</i>	3.062	0.167	0.0000***	0.1000	—	<i>I</i> (0)
	<i>l_ESCAPE_Index</i>	3.062	0.167	0.0003***	0.1000	—	<i>I</i> (0)
	<i>l_INFL_sa</i>	0.072	0.071	0.0020***	0.1000	—	<i>I</i> (0)
	<i>l_IIP_sa</i>	0.9575	0.0100	0.9575	0.0100	0.0000***	<i>I</i> (1)
	<i>l_EXR</i>	0.5006	0.0100	0.5006	0.0100	0.0000***	<i>I</i> (1)

Note: ***, **, and * denote statistical significance at the 1%, 5%, and 10% p-levels, respectively. All estimations are compiled with a restricted deterministic constant (*c*). For *l_INFL_sa* in the US system, the conflicting Augmented Dickey-Fuller (ADF) ($p = 0.0001$) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) ($p = 0.0377$) configurations indicate fractional integration induced by a structural break baseline, managed downstream via heteroskedasticity and autocorrelation consistent (HAC) standard errors.

The baseline parameters reported in Table 1 formally secure the unit root properties observed in the visual trajectories. The ADF and KPSS diagnostics confirm that in the United States, all primary behavioral metrics, including *l_SHOCK_Index* ($p = 0.6678$), *l_VOL_Base_Index* ($p = 0.1674$), *l_MDI_Index* ($p = 0.6925$), and *l_ESCAPE_Index* ($p = 0.7420$), consistently fail to reject the null hypothesis of a unit root at level while rejecting the KPSS stationarity null ($p = 0.0100$). This verifies that these series operate as non-stationary integrated processes of order one (*I*(1)) and resolve into stationary strings only upon first-differencing (Δ^1).

In stark opposition, the Vietnamese macroeconomic matrix displays a mixed integrated profile: while *l_SHOCK_Index* exhibits strict non-stationarity (*I*(1), $p = 0.9092$), aggregate *l_VOL_Base_Index* ($p = 0.0000$), *l_MDI_Index* ($p = 0.0000$), and *l_ESCAPE_Index* ($p = 0.0003$) are deeply stationary at level (*I*(0)) with robust mean-reverting properties. This statistical configuration confirms the mean-reverting behavior visualized in Figure 3, validating that collective anxiety acts as a transient phenomenon within the Vietnamese cultural architecture. Crucially, neither market displays dangerous *I*(2) explosive roots,

validating the technical boundaries required for stable macro-econometric modeling.

3.3 Analytical Strategy and Econometric Specification

Diverging from static cross-lagged survey panels, we formalize a dynamic macro-econometric modeling pipeline optimized for high-frequency tracking.

Dynamic Uncertainty Filters: Integrated Generalized Autoregressive Conditional Heteroskedasticity (IGARCH) vs. Exponentially Weighted Moving Average (EWMA) To construct the continuous uncertainty proxy (AI_{Vol}) dictated by the ARCH-LM parameters, the pipeline dynamically splits its filtering logic. For the high-heteroskedasticity US system, the conditional variance exhibits high persistence ($\alpha + \beta \approx 1.000$), which we capture via an IGARCH framework:

$$\sigma_{t,US}^2 = \omega + \alpha \xi_{t-1}^2 + (1 - \alpha) \sigma_{t-1}^2, \quad (2)$$

where σ_t^2 maps the evolution of conditional variance as the primary AI_{Vol} series. For the Vietnamese system, due to the homoskedastic baseline of key indicators, an EWMA with an optimized decay parameter ($\lambda = 0.94$) is deployed to extract clean, short-memory volatility movements:

$$\sigma_{t,VN}^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) \xi_{t-1}^2. \quad (3)$$

SVAR and Identification Restrictons To untangle the concurrent interactions and capture causal lags, we estimate a SVAR system. Let $X_t = [AI_{Vol}, GenAI_{Shock}, Job_{Insecurity}, Burnout, Job_{Quitting}]'$ represent the endogenous vector. The structural model is formulated as:

$$A_0 X_t = c + \sum_{j=1}^p A_j X_{t-j} + \varepsilon_t, \quad (4)$$

where A_0 is the matrix of contemporaneous parameters, A_j denotes the historical lag matrices optimized at $p = 4$ weeks, and ε_t represents the vector of orthogonalized structural shocks.

To model the behavioral latency required for internal cognitive processing and resource depletion, we specify contemporary zero-restrictions on the A_0 matrix instead of a standard recursive Cholesky ordering:

$$A_0 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ a_{21} & 1 & 0 & 0 & 0 \\ a_{31} & a_{32} & 1 & 0 & 0 \\ 0 & a_{42} & a_{43} & 1 & 0 \\ 0 & a_{52} & a_{53} & a_{54} & 1 \end{bmatrix}. \quad (5)$$

By setting $a_{41} = 0$ and $a_{51} = 0$, the structural architecture prevents instantaneous, same-week pass-through from technological volatility shocks to collective burnout and quitting, forcing the transmission to propagate systematically through chronological historical lag blocks ($t - 1$ to $t - 4$). This structural ordering strictly adheres to the established psychological sequence of occupational distress: cognitive appraisal (*Job Insecurity*) precedes affective strain (*Burnout*), which ultimately triggers final behavioral withdrawal (*Job Quitting*)¹.

Regime Shifts and Predictive Validation To capture structural slope accelerations across the post-ChatGPT threshold, we specify a regime-dependent *Difference-in-VAR* interaction framework:

$$\Delta Y_t = \alpha + \beta_{\text{base}} \Delta G_t + \beta_{\text{inter}} (\Delta G_t \times D_t) + \gamma D_t + \theta \widehat{EC}_{t-1} + v_t, \quad (6)$$

where D_t is a binary threshold indicator maximizing at 1 after November 2022, and β_{inter} explicitly isolates the behavioral acceleration.

To safeguard the estimates against overfitting and sample-size distortions in post-break windows, the econometric outputs are cross-validated against a shallow, regularized Random Forest engine ($\text{max_depth} = 4$). Crucially, by feeding first-differenced, stationary data (ΔY) into the Random Forest algorithm, we actively prevent the classical extrapolation failure commonly associated with tree-based models when applied to trending time-series data. Out-of-sample forecasting consistency is formally verified via scale-stationary Diebold-Mariano (DM) tests.

4 Empirical Results

As an initial step in modeling high-frequency systemic shifts following the widespread commercialization of GenAI, we examined the structural behavior of labor market metrics across two contrasting national economic frameworks: the United States and Vietnam. To ensure statistical validity prior to full structural estimation, baseline diagnostic pipelines including asset stationarity screening, optimal lag selection, and residual white-noise verifications were rigorously executed.

For the United States system, the vector autoregressive model was conditionalized on an optimal lag vector of $p = 1$ based on the strict prioritization of the Schwarz Information Criterion (BIC). To eliminate potential higher-order autocorrelation and account for a more complex lagged feedback structure in an emerging economic matrix, the Vietnamese system was parameterized at a deeper specification of $p = 4$, successfully clearing the multivariate Lagrange Multiplier diagnostic framework (MV-LM $p = 0.171$).

¹ To ensure econometric robustness, we estimated alternative SVAR specifications with inverted orderings (e.g., placing Job Quitting before Burnout). The resulting Impulse Response Functions (IRF) and Forecast Error Variance Decomposition (FEVD) remained structurally consistent, confirming that our findings are not artifacts of the chosen Cholesky factorization.

4.1 Short-Run Innovations and Systemic Variance Splits

To address the directional predictive primacy within these structural frameworks, we formalized multivariate Granger causality tests before examining contemporaneous shock vectors. Table 2 summarizes the causal trajectories mapped between the exogenous technological diffusion signals and localized collective labor responses.

Table 2: Multivariate VAR Granger Causality Testing Matrix

Directional Predictive Trajectory	F -Stat	p -value	Decision
United States Model ($p = 1$)			
$\Delta\text{GenAI Shock} \rightarrow \Delta\text{Job Insecurity}$	5.603	0.004***	Reject H_0 (Causal Path Active)
$\Delta\text{Job Insecurity} \rightarrow \Delta\text{GenAI Shock}$	0.969	0.380	Fail to Reject H_0 (Unidirectional)
Vietnam Model ($p = 4$)			
$\Delta\text{GenAI Shock} \rightarrow \text{Job Insecurity (Level)}$	2.921	0.055*	Reject H_0 (Marginally Active)
$\text{Job Insecurity (Level)} \rightarrow \Delta\text{GenAI Shock}$	6.641	0.001***	Reject H_0 (Highly Feedback-Active)

Note: ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Regarding the causal structures illustrated in Table 2, we noted a sharp, fundamental divergence between the two economic regimes. In the United States, a powerful, unidirectional predictive link was observed running exclusively from technological volatility to labor market anxiety. The null hypothesis that generative automation shock waves do not Granger-cause workforce job insecurity was robustly rejected ($F = 5.603, p = 0.004$). Crucially, the reverse feedback loop from aggregated labor anxiety back to technological diffusion failed to meet any meaningful significance threshold ($F = 0.969, p = 0.380$). This suggests that in an individualistic labor ecosystem, market-wide salience of generative technology operates as a purely exogenous cognitive disruptor.

Conversely, the Vietnamese market exhibited a completely reversed behavioral reality. The feedback vector running from domestic job insecurity back into technological search trajectories was highly significant and statistically dominant ($F = 6.641, p = 0.001$), whereas the direct transmission from the exogenous shock to localized insecurity was heavily muted, remaining only marginally active at the 10% boundary ($F = 2.921, p = 0.055$).

To isolate how these innovations cascade through the entire psychological and behavioral spectrum over a half-year operational horizon, we computed structural FEVDs at a 26-week forecast boundary. Table 3 details the comparative

allocation of system variance generated via a data-driven Cholesky ordering identification.

Table 3: Structural Forecast Error Variance Decomposition (FEVD) at Week 26 Horizon

Decomposition Target	(1)	(2)	(3)	(4)	(5)
United States					
(1) AI Volatility	4.74%	38.37%	7.65%	16.45%	32.78%
(2) GenAI Shock	4.61%	38.81%	7.32%	16.49%	32.77%
(3) Job Insecurity	4.67%	38.78%	7.30%	16.48%	32.77%
(4) Burnout (MDI)	4.68%	38.77%	7.30%	16.48%	32.77%
(5) Job Quitting	4.69%	38.77%	7.30%	16.47%	32.76%
Vietnam					
(1) AI Volatility	5.55%	10.09%	4.38%	13.83%	66.16%
(2) GenAI Shock	4.58%	21.58%	6.65%	14.69%	52.50%
(3) Job Insecurity	5.31%	20.13%	6.44%	14.08%	54.04%
(4) Burnout (MDI)	7.10%	18.41%	5.93%	14.02%	54.55%
(5) Job Quitting	9.78%	17.39%	5.57%	13.88%	53.39%

Note: (1) AI Volatility; (2) GenAI Shock; (3) Job Insecurity; (4) Burnout (MDI); (5) Job Quitting. Variance shares are expressed in percentages (%).

The explanatory power of the structural innovations at the 26-week level of analysis was remarkable, revealing two distinct structural archetypes. In the United States, we noted an extreme compression of variance alignment; the explicit generative shock component stabilized at a massive, uniform threshold of approximately 38.77% to 38.81% across all internal targets. Concurrently, the ultimate behavioral exit coordinate (Job Quitting) locked into a tight variance split of 32.76% to 32.78%. This absolute synchronization points to a highly fluid, frictionless distress transmission structure within individualistic labor markets.

In contrast, the Vietnamese variance distribution was heavily dominated by its own internal historical memory. For the primary behavioral withdrawal metric (Job Quitting), its own structural historical block retained a clear majority of 53.39%, while the cumulative explanatory footprint of the technological innovation shock (GenAI Shock) was heavily constrained to 17.39%. Similarly, the direct transmission of technological shocks toward collective employment panic (Job Insecurity) was restricted to 20.13%, validating the presence of robust structural insulation or buffering properties within the emerging East Asian workforce.

4.2 Regime-Switching Dynamics and Structural Break Analysis

To address the role of explicit technological shocks over historical time, we examined whether the commercial release of ChatGPT in November 2022 generated

a systematic, non-linear structural break in parameter distribution. Splitting the sample into a Pre-ChatGPT window ($N = 46$) and a Post-ChatGPT window ($N = 170$), we deployed a split-sample ARDL Bounds framework coupled with a formal Chow structural stability test. Table 4 summarizes these long-run cointegrating dynamics and stability diagnostics.

Table 4: Split-Sample ARDL Bounds Testing and Chow Diagnostics

Market & Model	Period	ARDL	F -Stat	Verdict	Speed (ϕ)	p	Chow F
United States							
<i>M1: Job Insecurity</i>	Pre	(2,1)	0.674	No Coint.	N/A	N/A	7.597***
	Post	(1,1)	22.745	Coint.	-0.318	0.000	
<i>M2: Burnout</i>	Pre	(1,1)	3.970	Inconcl.	-0.445	0.003	3.200***
	Post	(1,1)	56.963	Coint.	-0.111	-0.039	
<i>M3: Job Quitting</i>	Pre	(2,1)	3.228	Inconcl.	-0.662	0.001	1.808**
	Post	(1,1)	55.958	Coint.	-0.279	0.000	
Vietnam							
<i>M1: Job Insecurity</i>	Pre	(1,1)	0.600	No Coint.	N/A	N/A	2.462***
	Post	(2,1)	4.625	Coint.	-0.367	0.000	
<i>M2: Burnout</i>	Pre	(1,1)	3.671	Inconcl.	-0.888	0.000	2.691***
	Post	(1,1)	17.708	Coint.	-0.620	0.000	
<i>M3: Job Quitting</i>	Pre	(1,1)	3.305	Inconcl.	-0.608	0.010	2.225***
	Post	(3,1)	14.893	Coint.	-0.316	0.000	

Note: *Coint.* = Cointegrated; *Inconcl.* = Inconclusive. Critical thresholds follow Pesaran et al. (2001). ***, **, and * denote significance at 1%, 5%, and 10%.

The empirical evidence documented in Table 4 demonstrates a profound regime shift in global labor market mechanics. The primary null hypothesis of parameter stability across the November 2022 temporal boundary was uniformly rejected by the Chow test across all estimated models for both economies.

In the United States, the structural break was especially pronounced for Job Insecurity ($F = 7.597, p < 0.001$) and Burnout ($F = 3.200, p < 0.001$). Prior to the ChatGPT release, long-run cointegration between generative automation signals and labor market anxiety was statistically absent. However, in the post-break regime, these systems locked into highly significant cointegrating trajectories, with the bounds F -statistic surging to 22.745 for job insecurity and 56.963 for burnout, vastly exceeding the Pesaran asymptotic upper bounds.

To trace the precise nature of these parameters, we introduced an interactive time-dummy framework ($\Delta \text{GenAI Shock} \times D_t$). In the US Burnout model ($M2$), the interaction coefficient was positive and statistically explosive ($\beta_{\text{inter}} = +0.606, p < 0.001$). This suggests that the transmission slope fundamentally flipped from a non-significant baseline in the pre-era ($\beta_{\text{pre}} = -0.016$) to an intensely positive slope post-ChatGPT ($\beta_{\text{post}} = +0.590$), confirming that the

mass public availability of generative tools shifted the technology from an abstract concept into an active corporate stressor.

Concurrently, the Vietnamese market experienced a highly sophisticated, inverse parameter realignment. The Chow diagnostics heavily rejected parameter stability ($M2_Burnout$ Chow $F = 2.691, p = 0.002$). Strikingly, the interaction model captured a massive, negative slope shift ($\beta_{inter} = -1.718, p < 0.001$). This systematic intervention caused the direct transmission elasticity to down-scale drastically from an elevated baseline in the pre-era ($\beta_{pre} = +2.190$) down to a muted long-run elasticity post-ChatGPT ($\beta_{post} = +0.473$). This suggests that while the initial viral wave of GenAI sparked conversational panic among Vietnamese labor cohorts, the post-break integration led to rapid psychological adaptation and normalization within operational routines.

4.3 Predictive Validation and Robustness Frameworks

To protect our structural inferences against potential overfitting or linear model specification bias, we implemented an out-of-sample predictive benchmark. A non-linear, constrained Random Forest algorithm ($max_depth = 4$) was trained on an identical first-differenced training footprint ($N_{train} = 137$) and evaluated against a purely blind forward test block ($N_{test} = 35$). Table 5 presents the out-of-sample Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), validated by the DM historical accuracy test against a driftless Naive benchmark.

Table 5: Out-of-Sample Predictive Errors and Diebold-Mariano Benchmarks

Target Coordinate	Specification	RMSE	MAE	Dir. Acc.	DM Test (p)
United States					
Δ Job Insecurity	Error Correction Model (ECM)-OLS	0.186	0.135	65.71%	2.339 (0.019)**
	Random Forest	0.140	0.105	54.29%	-0.272 (0.786)**
Δ Burnout (MDI)	ECM-OLS	0.219	0.160	51.43%	2.441 (0.015)**
	Random Forest	0.210	0.140	62.86%	2.042 (0.041)**
Vietnam					
Δ Job Insecurity	ECM-OLS	0.746	0.534	74.29%	-2.732 (0.006)***
	Random Forest	0.835	0.596	71.43%	-2.605 (0.009)***
Δ Burnout (MDI)	ECM-OLS	0.208	0.150	51.43%	-0.068 (0.946)**
	Random Forest	0.209	0.145	62.86%	-0.046 (0.963)**

Note: *Dir. Acc.* = Directional Accuracy. RMSE and MAE are evaluated directly on the Δy scale. ***, **, and * denote significance at 1%, 5%, and 10%.

The predictive diagnostic landscape reported in Table 5 establishes the statistical robustness of the empirical system. For the primary affective strain variable (Δ Burnout), the machine learning architecture achieved substantial predictive

optimization. In the US market, the Random Forest model successfully drove forecasting errors down to an RMSE of 0.210 and an MAE of 0.140, outperforming the OLS specification while securing an impressive directional accuracy threshold of 62.86%. Crucially, the DM test validated that this forecasting edge over the baseline was statistically significant ($DM = 2.042, p = 0.041$).

For the Vietnamese market, the predictive infrastructure highlighted the exceptional efficiency of the error-correction model in capturing localized labor panic. The Vietnamese ECM-OLS model for Job Insecurity delivered a dominant directional accuracy of 74.29%, outperforming both the Random Forest model (71.43%) and the naive baseline. The superiority of these error-correction structural features was forcefully confirmed by the DM metric ($DM = -2.732, p = 0.006$). This predictive verification demonstrates that the economic transmission paths mapped by our structural equations are not artifacts of statistical overfitting; rather, the inclusion of cointegrating error-correction vectors and localized GARCH volatility coordinates yields real, quantifiable predictive power.

5 Discussion

Our findings are in some ways consistent and in other ways inconsistent with traditional economic theories and research on reciprocal relations between technological shocks and labor market welfare. By synthesizing these multi-model econometric outputs, we advance a comprehensive theoretical framework that reconciles quantitative macro-vulnerabilities with macro-organizational psychological theories.

5.1 Psychological Resource Depletion and the JD-R Framework

The continuous tracking lag and systemic variance patterns captured in our SVAR and ARDL architectures provide powerful empirical expansions for the JD-R model [7] under macrotechnological disruption. Historically, JD-R applications have operated primarily within static, micro-level organizational configurations. Our findings mathematically elevate these concepts to dynamic macro-market frequencies.

Generative artificial intelligence shock functions serve as an unmitigated, structural *Job Demand*. When a technological breakthrough achieves market-wide salience, it introduces acute job demands in the form of required rapid cognitive adaptation, widespread restructuring, and a persistent threat of automation. As established by the US Granger causal vector ($\Delta\text{GenAI Shock} \rightarrow \Delta\text{Job Insecurity}, p = 0.004$), this demand does not immediately spark physical displacement; instead, it initiates an intense cognitive load of chronic career ambiguity.

According to the JD-R health impairment process, prolonged exposure to high job demands without adequate supporting resources systematically drains an employee’s energy, triggering chronic professional burnout. Our split-sample interaction models provide definitive proof of this energy erosion mechanism: the

sudden post-ChatGPT structural acceleration in the US burnout vector ($\beta_{\text{post}} = +0.590, p < 0.001$) reveals a population undergoing severe psychological resource depletion.

Once personal coping resources are completely exhausted, the behavioral withdrawal process is activated as an ultimate self-preservation response. The massive shared variance structure uncovered in the US FEVD table (where GenAI and Job Quitting lock together at $\sim 32\%$ and $\sim 38\%$ thresholds) maps this exact progressive path: macro-technological disruptions generate structural psychological stress, which rapidly erodes individual cognitive reserves, leading inexorably to systematic behavioral withdrawal from the labor market.

5.2 Cultural Buffering Mechanisms: Individualism vs. Collectivism

The stark structural differences exposed between the United States and Vietnamese models provide compelling validation of the cultural cushioning hypothesis within international labor markets. By mapping these economic equations against Hofstede’s (2011) cultural dimensions, we expose how underlying social values completely reshape how workers process technological anxiety.

The United States represents an intensely individualistic culture (Individualism score = 91), characterized by fluid employment relationships, minimal institutional job protection, and decentralized welfare frameworks. In this environment, when a major technological disruption occurs, the workforce processes it through a hyper-localized, individualistic lens of survival. Every automation signal directly threatens the individual’s market value, an effect severely exacerbated by the “at-will” employment doctrine and the synchronous 2022-2023 “Tech Winter” layoffs in the US labor market. Because there are no strong social cushioning mechanisms to absorb the blow, this individualistic anxiety immediately translates into severe burnout and rapid voluntary exit behavior. This vulnerability is clearly reflected in the US FEVD matrix, where external technological shocks seamlessly fracture the entire labor landscape.

In direct contrast, Vietnam operates under a deeply collectivist paradigm (Individualism score = 20), supported by strong organizational mutual obligations, tight-knit family safety nets, and institutional expectations of employment stability. When the generative AI wave hit Vietnam, it initially provoked massive conversational anxiety, as shown by the high pre-break elasticity ($\beta_{\text{pre}} = +2.190, p < 0.001$). However, our post-ChatGPT interaction models capture an extraordinary adaptive downscaling ($\beta_{\text{post}} = +0.473, p < 0.001$).

This empirical deceleration indicates that collectivist networks act as powerful shock absorbers or psychological “buffers” that intercept and neutralize technological stress. However, it is essential to acknowledge that this buffering effect is concurrently reinforced by structural economic conditions. Vietnam’s lower labor costs, higher institutional barriers to firing, and delayed integration of core AI technologies relative to the US provided an inherent temporal and economic cushion. Thus, rather than resorting to individualistic exit strategies (Job Quitting), the workforce leverages both social support and institutional lags to actively integrate and normalize the technology, preserving structural

continuity and insulating the broader national labor market from systemic attrition.

5.3 Theoretical and Practical Implications

Overall, these findings suggest that the relation between macro-technological innovations and employee welfare is highly complex, non-linear, and intensely moderated by cultural architecture. This empirical finding contributes to the continued development of dynamic models of stress processes (e.g., Guthier et al., 2020), especially those that propose cyclical and cumulative processes, such as the conservation of resources theory (Hobfoll, 1989). While traditional micro-organizational research has long documented the negative impacts of change anxiety, our study provides a structural macro-economic template for capturing the transmission speeds and boundary limits of these shocks.

Practically, our findings should alert policymakers, considering the simultaneous rise of GenAI deployment and workforce mental health concerns globally. Public health officials must begin to view corporate technological transition strategies as major factors in occupational welfare. Our data supplement public health frameworks by providing high-frequency insights into how these linkages occur. Interventions are needed to target these vulnerabilities, combining corporate transparency and multi-level supervisor support systems to buffer employees against the psychological resource depletion associated with rapid technological displacement.

5.4 Limitations and Future Research

Despite the analytical strengths of using high-frequency, multi-country macro data, several limitations must be noted. First, our proxies for labor market reactions, while granular, rely heavily on digitized search behavior and behavioral sentiment indices. Future research should look to complement these high-frequency macro trends with matched longitudinal panel surveys across organizations to verify the micro-foundations of the cultural buffer effects. Furthermore, while we emphasize cultural dimensions, future research must disentangle collectivist networks from overlapping macroeconomic structures (e.g., structural labor costs, employment legislation, and tech-sector maturity) to isolate the precise weight of each buffering mechanism. Second, while our stationary Random Forest architecture confirmed the presence of non-linear forecasting capacities, the operational mechanisms remain somewhat black-boxed compared to the structural ARDL parameters. Future inquiries would benefit from deploying interpretable machine learning approaches (such as SHAP values) over long-horizon macro data to fully map the evolving structural boundaries of technological stress.

6 Conclusion and Practical Implications

6.1 Summary of Empirical Findings

This study provided a high-frequency, cross-national examination of how macro-technological innovations—specifically the commercialization and diffusion of GenAI—systematically reshape labor market welfare, professional burnout, and behavioral withdrawal intentions. By contrasting the structural dynamics of the United States and Vietnam, we illuminated how underlying cultural architectures alter the transmission mechanisms of technological anxiety.

Our empirical findings yield several critical conclusions. First, SVAR diagnostics demonstrated that the market-wide salience of GenAI operates as an intense, exogenous shock in individualistic settings. In the United States, we observed a strict unidirectional causal path ($\Delta\text{GenAI Shock} \rightarrow \Delta\text{Job Insecurity}$, $p = 0.004$), where continuous innovations rapidly locked into a highly synchronized variance infrastructure, accounting for nearly 38.77% of the systemic variance in workforce burnout and voluntary exit behavior (Job Quitting).

Second, our split-sample ARDL bounds frameworks and formal Chow diagnostics confirmed that the public launch of ChatGPT in November 2022 marked a non-linear structural break globally. In the United States, this transition initiated an acute resource depletion process, causing the long-run transmission elasticity to accelerate sharply ($\beta_{\text{post}} = +0.590, p < 0.001$). Conversely, the Vietnamese market demonstrated a unique cultural cushioning mechanism. Supported by collective organizational safety nets, the Vietnamese workforce successfully downscaled long-run shock elasticities over the post-ChatGPT regime ($\beta_{\text{post}} = +0.473, p < 0.001$), absorbing external technological stress through social normalization and proactive, defensive information-seeking trajectories ($F = 6.641, p = 0.001$).

6.2 Practical and Policy Implications

The reciprocal and intensifying dynamics observed over time, particularly regarding the severe psychological resource depletion pathways, yield profound practical implications for public policymakers, public health officials, and corporate leaders. Because “health and employment are inextricably linked,” organizations can no longer view technological restructuring purely through the lens of operational efficiency; instead, macro-technological transitions must be actively managed as primary determinants of occupational health.

For corporate managers, especially within fluid, individualistic labor markets, our findings alert leaders to the high risk of systematic workforce attrition following aggressive AI integration. To prevent chronic professional burnout, organizations must combine individual- and organizational-level interventions. Management should prioritize transparent communication protocols and participatory restructuring frameworks to reduce job ambiguity and career insecurity. Furthermore, supervisor social support and targeted upskilling initiatives must

be deployed as crucial organizational resources to buffer employees against cognitive resource depletion.

For public policymakers and health officials, these findings reinforce the necessity of institutional protection frameworks during periods of rapid macroeconomic dislocation. Governments should invest in continuous digital literacy infrastructures and regional psychological counseling resources. By treating stable, psychologically secure employment as a critical contributor to public health, policymakers can design multi-level intervention strategies that protect structural continuity, buoy workforce resilience, and prevent large-scale societal burnout as automated technologies advance.

6.3 Concluding Remarks

In conclusion, the intersection of GenAI and global labor dynamics represents one of the defining structural transformations of the modern era. This research successfully moves beyond static, micro-level models of workplace stress by establishing a dynamic macro-economic template to capture the transmission boundaries of technological anxiety. While individualistic environments exhibit a high structural vulnerability to unmitigated technological demands, collectivist networks provide a powerful psychological buffer that breaks the mechanical link between conversational panic and behavioral exit. Ultimately, sustaining workforce welfare in the age of automation requires a conscious shift in corporate philosophy, one that balances technological acceleration with the preservation of human cognitive and social resources.

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